

Kmos-0205

SUMMARY

Version: 1.0 Status: Draft

PURPOSE

The Capability Registry Service is the authoritative catalog of executable business capabilities within the Knowledge & Media Operating System (KMOS).

Its responsibility is to discover, register, classify, certify, version, govern, and expose the business abilities available to the platform.

The Capability Registry SHALL answer questions such as: - What can the platform do? - Which Capability performs this function? - Which version should be executed? - Is the Capability certified? - What are its dependencies? - What business contracts does it implement?

The Capability Registry SHALL NOT be a network service registry.

It is the institutional catalog of execution knowledge.

MISSION

Provide a permanent, governed inventory of every business Capability available to KMOS.

The Capability Registry answers the question: "What institutional abilities are available to perform this work?"

RESPONSIBILITIES

The Capability Registry SHALL own: Capability Registration Capability Manifest Capability Metadata Capability Version History Capability Classification Capability Lifecycle Capability Certification Capability Dependencies Capability Compatibility Capability Discovery Capability Contracts Capability Documentation Capability Health Metadata Capability Policies Capability Registry Events

NON-RESPONSIBILITIES

The Capability Registry SHALL NOT: Execute Capabilities Coordinate Workflows Store Knowledge Objects Store Assets Manage Infrastructure Schedule Jobs Perform Business Processing Maintain Network Service Discovery

Execution belongs to the Capability Runtime. Coordination belongs to the Workflow Service.

CANONICAL OBJECTS OWNED

The Capability Registry SHALL be the authoritative owner of: Capability Capability Manifest Capability Version Capability Contract Capability Certification Capability Classification Capability Dependency Capability Profile Capability Documentation Capability Lifecycle Record

Future Capability objects SHALL extend KMOS-0120 and KMOS-0160.

ENGINEERING PHILOSOPHY

Capabilities describe business abilities. Implementations realize those abilities.

Examples: Speech Recognition Transcript Correction Vocabulary Expansion Knowledge Extraction Story Generation Publishing Rendering Translation

The Capability Registry catalogs business meaning rather than software deployment.

INTERNAL MODULES

The reference implementation SHOULD contain: Capability Registry Manifest Manager Contract Manager Version Manager Dependency Manager Certification Manager Compatibility Manager Discovery Engine Documentation Manager Health Metadata Manager Policy Manager Capability API Capability Event Publisher

Each module SHALL have one clearly defined responsibility.

PUBLIC BUSINESS APIS

The Capability Registry SHALL expose business APIs. Examples include: Register Capability Retrieve Capability Search Capabilities Retrieve Capability Contract Retrieve Capability Manifest Retrieve Capability Versions Retrieve Dependencies Retrieve Certification Retrieve Compatibility Approve Capability Deprecate Capability Retire Capability Validate Capability Package Export Capability Metadata Import Capability Metadata

Business APIs SHALL expose business abilities rather than runtime infrastructure.

CANONICAL EVENTS PUBLISHED

Examples include: CapabilityRegistered CapabilityUpdated CapabilityCertified CapabilityRejected CapabilityDeprecated CapabilityRetired CapabilityVersionCreated CapabilityCompatibilityChanged CapabilityHealthChanged CapabilityDocumentationUpdated ManifestValidated DependencyUpdated

Every lifecycle transition SHALL publish canonical events.

CANONICAL EVENTS CONSUMED

Examples include: WorkflowRegistered ExtensionInstalled ExtensionRemoved CertificationGranted CertificationRevoked ConfigurationChanged CapabilityPackagePublished CapabilityRuntimeAvailable CapabilityRuntimeUnavailable

These events SHALL update registry state.

CAPABILITY MANIFEST

Every registered Capability SHALL include a canonical manifest. The manifest SHALL contain: Canonical Identifier Business Name Business Purpose Owner Domain Supported Inputs Supported Outputs Consumed Events Published Events Configuration Schema Security Requirements Dependencies Certification Level Documentation References Version Information

The manifest SHALL be the authoritative description of the Capability.

CAPABILITY DISCOVERY

Discovery SHALL support searching by: Business Function Domain Input Type Output Type Knowledge Type Asset Type Workflow Compatibility Certification Level Lifecycle State Language Organization

Capability discovery SHALL remain independent of runtime topology.

CAPABILITY CONTRACTS

Every Capability SHALL expose a canonical business contract. The contract SHALL define: Business Purpose Accepted Objects Produced Objects Published Events Consumed Events Preconditions Postconditions Business Constraints Error Conditions

Contracts SHALL remain stable across implementation changes.

DEPENDENCY MANAGEMENT

The Registry SHALL maintain explicit dependency graphs. Dependency categories include: Capabilities Knowledge Models Policies Configuration External Connectors Extensions Platform Services

Circular dependencies SHALL be detected and rejected unless explicitly supported by a defined execution pattern.

CERTIFICATION

Certification SHALL evaluate: Business Compliance Contract Compatibility Security Performance Governance Documentation Testing Operational Readiness

Certification SHALL be independent of deployment environment.

VERSION MANAGEMENT

Capabilities SHALL support immutable version history. Each version SHALL record: Version Identifier Publication Date Business Changes Compatibility Dependencies Certification Approval Deprecation Status

Historical versions SHALL remain discoverable.

COMPATIBILITY

Compatibility SHALL be evaluated against: KMOS Version Workflow Definitions Knowledge Schemas Asset Schemas Event Schemas Extension Versions API Profiles

Compatibility SHALL be explicit rather than assumed.

HEALTH METADATA

The Registry SHALL record operational metadata. Examples include: Current Status Availability Last Validation Supported Runtime Supported Platform Version Operational Notes

The Registry SHALL record metadata rather than monitor infrastructure directly.

SECURITY

The Capability Registry SHALL support: Authentication Authorization Capability Signing Manifest Validation Audit Logging Certification Verification Policy Enforcement

Security SHALL conform to KMOS-0190.

OBSERVABILITY

The Registry SHALL publish: Registration Metrics Discovery Metrics Certification Metrics Version Statistics Dependency Metrics Compatibility Reports Lifecycle Statistics Operational Health Business Capability Metrics

SCALABILITY

The Capability Registry SHALL scale independently. Scaling dimensions include: Registered Capabilities Capability Versions Organizations Extensions Queries Certifications Dependencies Documentation

Scaling SHALL remain independent of runtime execution.

FAILURE STRATEGY

Failures SHALL never compromise institutional execution knowledge. Recovery SHALL support: Manifest Revalidation Registry Reconstruction Event Replay Audit Operational Alerts Backup Restoration

Registry integrity SHALL be continuously verifiable.

TESTING STRATEGY

Testing SHALL include: Registration Manifest Validation Contract Validation Dependency Resolution Compatibility Analysis Certification Workflow Version History Search Performance Migration Replay

ACCEPTANCE CRITERIA

A compliant Capability Registry SHALL: Maintain canonical Capability records. Maintain manifests. Maintain contracts. Support discovery. Support certification. Support dependency analysis. Publish canonical events. Support KMOS-0120. Support KMOS-0160. Remain runtime independent.

REALITY CHECK

Current implementation maturity.

Capability catalog | Media Pipeline capability libraries | Proven Modular processing pipeline | Media Pipeline | Proven Capability composition | AIMPOS | Proven Capability manifests | Partial implementation concepts | Partial Capability certification | Governance process | Partial Capability compatibility analysis | Planned | Early Dynamic discovery | Planned | Not Implemented Registry reconstruction | Planned | Not Implemented

The implementation SHALL prioritize extracting proven capability metadata before introducing advanced registry features.

REFERENCE IMPLEMENTATION

Suggested internal architecture.

capability-registry-service/ |—— api/ |—— application/ |—— domain/ | |—— registry/ | |—— manifests/ | |—— contracts/ | |—— versions/ | |—— certification/ | |—— compatibility/ | |—— discovery/ | |—— dependencies/ | |—— documentation/ |—— events/ |—— repositories/ |—— validators/ |—— tests/ |—— migrations/ |—— documentation/

The Registry SHALL remain independent of runtime execution engines.

CLAUDE IMPLEMENTATION GUIDE

Objective: Implement the authoritative Capability Registry Service.

Primary Specifications: KMOS-0120 KMOS-0160 KMOS-0170 KMOS-0200 KMOS-0204

Deliverables: Production-ready Capability Registry. Manifest Manager. Contract Manager. Discovery Engine. Dependency Analyzer. Certification Manager. Compatibility Engine. Automated Tests. Deployment Manifests. Operational Documentation.

Acceptance: The implementation SHALL satisfy every requirement defined in this specification. The Registry SHALL catalog business abilities rather than runtime infrastructure.

RELATIONSHIP TO OTHER SERVICES

Knowledge Service: Provides semantic classifications for Capabilities. Asset Registry: Stores Capability Packages and supporting artifacts. Event Service: Publishes and consumes Capability lifecycle events. Workflow Service: Discovers executable Capabilities. Identity Service: Authorizes Capability registration and administration. Governance Service: Certifies and audits Capabilities. Configuration Service: Provides runtime configuration profiles.

The Capability Registry SHALL remain the authoritative owner of institutional execution knowledge.

ARCHITECTURAL CONSEQUENCES

Because business abilities are centrally cataloged: Applications become dynamically composable. Workflows become implementation independent. Capabilities become reusable institutional assets. Extensions become discoverable. AI

models become interchangeable implementations. Platform evolution becomes additive rather than disruptive. Institutional execution knowledge survives technology replacement.

The Capability Registry becomes the institutional execution engine of KMOS.

REFERENCE EXAMPLE

A new "Transcript Correction" Capability is introduced. The Capability Package is submitted. Manifest is validated. Business contract is verified. Dependencies are analyzed. Compatibility is evaluated. Certification workflow is completed. Capability is registered. CapabilityRegistered event is published. Workflow Service discovers the new Capability. Future workflows may invoke it without any application changes.

The Capability becomes part of the organization's permanent execution knowledge.

NEXT SPECIFICATION

KMOS-0206 defines the Identity Service.

Where: The Knowledge Service preserves institutional meaning, The Asset Registry preserves institutional evidence, The Event Service preserves institutional communication, The Workflow Service preserves institutional coordination, The Capability Registry preserves institutional execution knowledge, the Identity Service preserves institutional accountability by establishing the canonical identities, authentication, authorization, organizational structure, and trust relationships required for every governed action within KMOS.

Identity becomes the foundation upon which accountability, governance, security, and institutional trust are built.